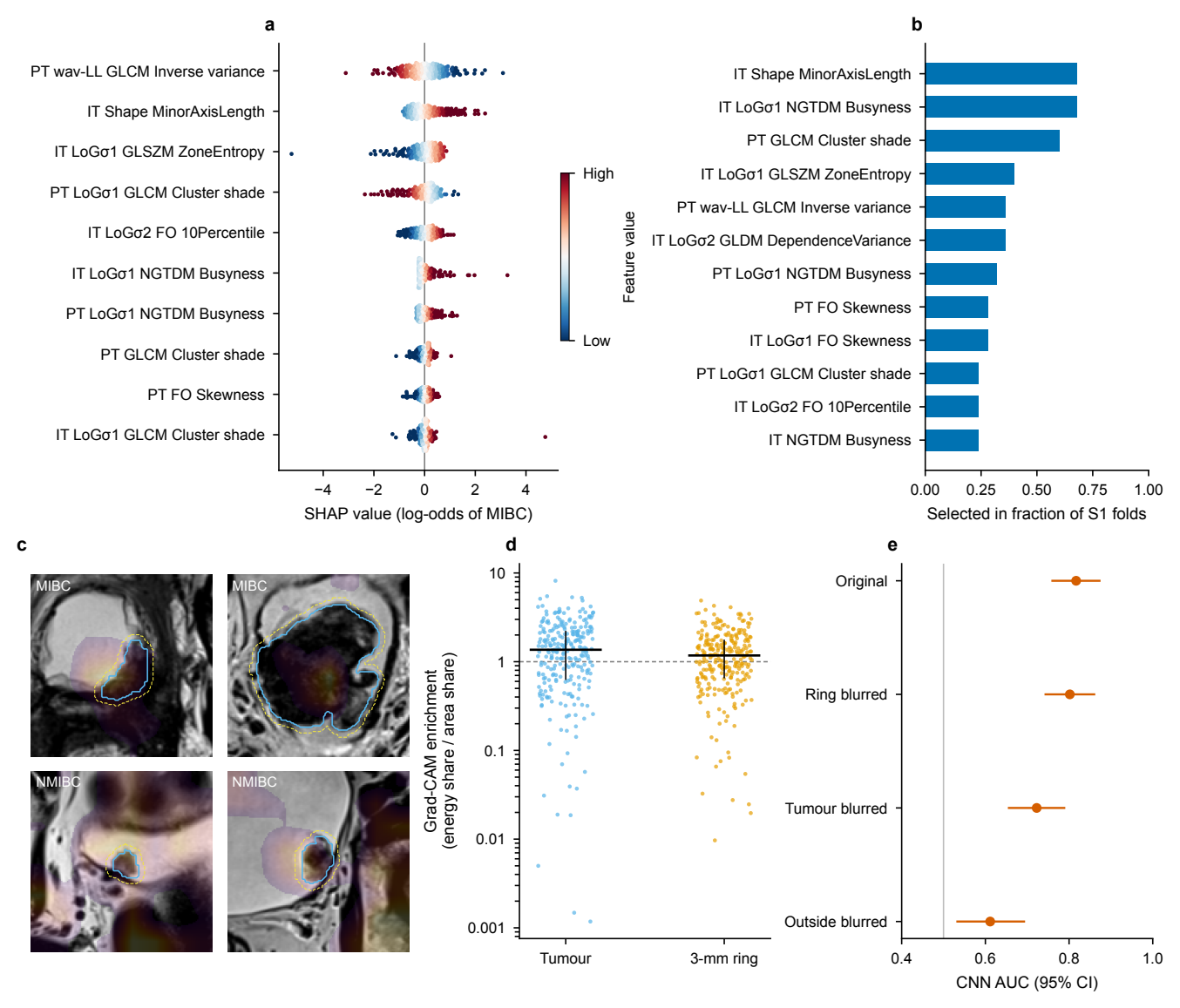

a
PT wav-LL GLCM Inverse variance
IT Shape MinorAxisLength
IT LoGσ1 GLSZM ZoneEntropy
PT LoGσ1 GLCM Cluster shade
IT LoGσ2 FO 10Percentile
IT LoGσ1 NGTDM Busyness
PT LoGσ1 NGTDM Busyness
PT GLCM Cluster shade
PT FO Skewness
IT LoGσ1 GLCM Cluster shade
−4
−2
0
2
4
SHAP value (log-odds of MIBC)
High
Low
Feature value
b
IT Shape MinorAxisLength
IT LoGσ1 NGTDM Busyness
PT GLCM Cluster shade
IT LoGσ1 GLSZM ZoneEntropy
PT wav-LL GLCM Inverse variance
IT LoGσ2 GLDM DependenceVariance
PT LoGσ1 NGTDM Busyness
PT FO Skewness
IT LoGσ1 FO Skewness
PT LoGσ1 GLCM Cluster shade
IT LoGσ2 FO 10Percentile
IT NGTDM Busyness
0.00
0.25
0.50
0.75
1.00
Selected in fraction of S1 folds
c
MIBC
MIBC
NMIBC
NMIBC
d
10
1
0.1
0.01
0.001
Grad-CAM enrichment
(energy share / area share)
Tumour
3-mm ring
e
Original
Ring blurred
Tumour blurred
Outside blurred
0.4
0.6
0.8
1.0
CNN AUC (95% CI)